

Postgres от начала веков до наших дней

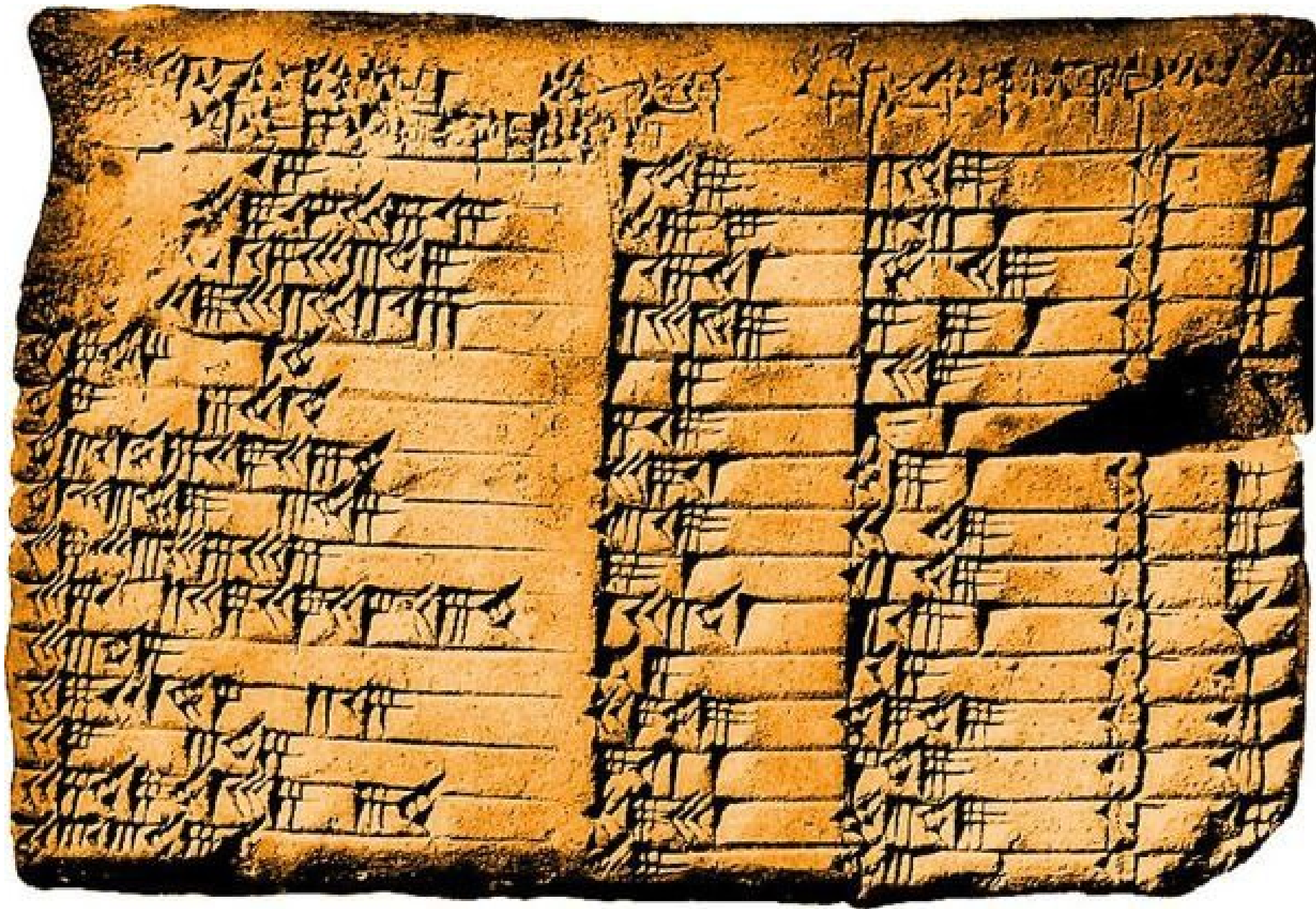
Олег Бартунов

Иван Панченко



**Saint
HighLoad⁺⁺**



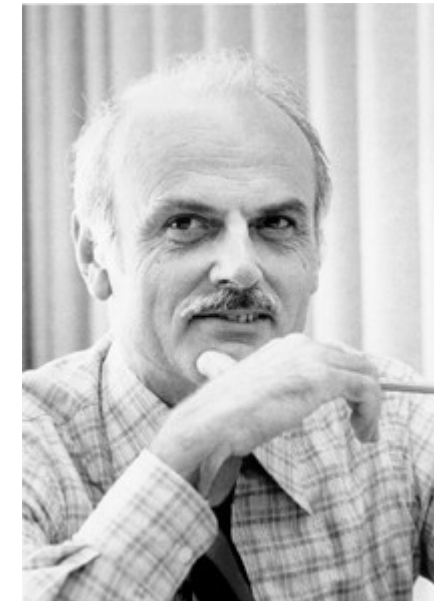


Были ли СУБД до СУБД ?

- Можно ли считать вавилонские таблички СУБД
- Хм, а что вообще мы понимаем под СУБД?
 - ACID
 - Структурированность данных
 - Оптимизация для поиска
- Определённые идеи, конечно, были
 - Таблицы :)
 - Картотеки :)
 - Двойная запись в бухгалтерии

У ОСНОВ СУБД ?

- Стаття Едгара Кодда:
- A relational model of data for large shared data banks. Communications of the ACM Volume **13**, 6 June 1970, pp 377–387
- <https://dl.acm.org/doi/10.1145/362384.362685>



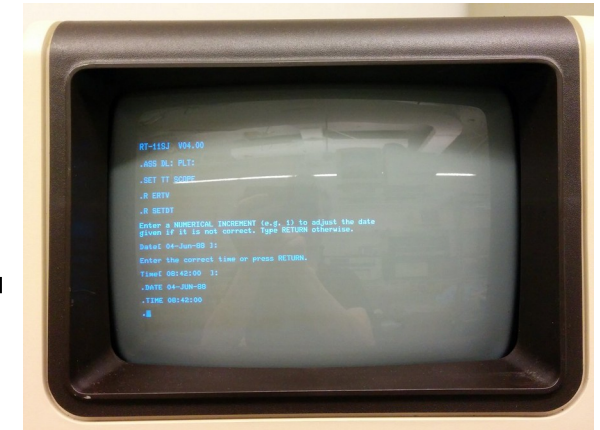
Ingres

- Чтобы разработать СУБД, Майкл Стоунбрейкер приобрел на грант пару «ПК» PDP-11.



- Когда подвезли терминал и поставили UNIX...
- Ему так понравилась интерактивная работа в shell, что он назвал СУБД

INteractive Graphic REtrieval System



Что было хорошего в Ingres

- Язык QUEL — почти SQL.
- Cost-based planner
- Data dictionary
- Access methods
- Locks
- Wei Hong набрал его код руками с распечатки в 1985 г.
- См <https://doi.org/10.1145/320473.320476>
<https://doi.org/10.1145/320141.320158>



Языки реляционных баз

- QUEL – QUERyLanguage
- SQUARE – Specifying Queries As Relational Expressions (1975)
Raymond Boyce,
Donald Chamberlin и др.
- PostQUEL
- SEQUEL - 1974
- SQL

SEQUEL: A STRUCTURED ENGLISH QUERY LANGUAGE

by

Donald D. Chamberlin
Raymond F. Boyce

IBM Research Laboratory
San Jose, California

ABSTRACT: In this paper we present the data manipulation facility for a structured English query language (SEQUEL) which can be used for accessing data in an integrated relational data base. Without resorting to the concepts of bound variables and quantifiers SEQUEL identifies a set of simple operations on tabular structures, which can be shown to be of equivalent power to the first order predicate calculus. A SEQUEL user is presented with a consistent set of keyword English templates which reflect how people use tables to obtain information. Moreover, the SEQUEL user is able to compose these basic templates in a structured manner in order to form more complex queries. SEQUEL is intended as a data base sublanguage for both the professional programmer and the more infrequent data base user.

Всё делали команды

Стоунбрейкер

- Лучшие ребята из Беркли
- QUEL — ошибка Стоунбрейкера.
- Она и убила Ingres (тогда)
- Академический и эксцентричный подход

Ларри Элинсон

- Лучшие ребята из Стэнфорда
- Ставка на SQL
- Который и стал стандартом
- Бизнес.

SQL queries run in this order

FROM + JOIN



WHERE



GROUP BY



HAVING



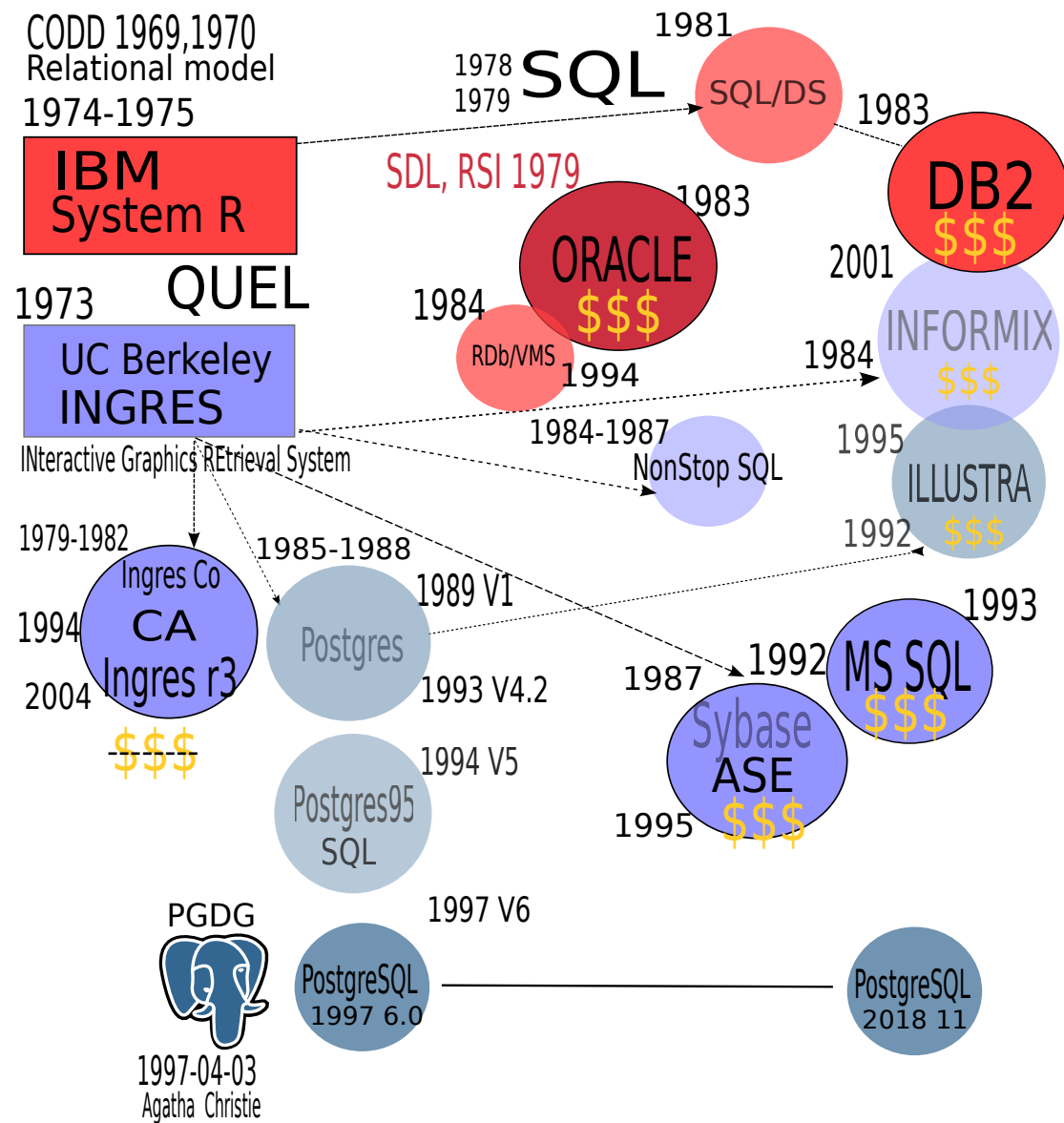
SELECT (window functions
happen here !)



ORDER BY



LIMIT



Postgres

- Stonebraker & Rowe. The design of Postgres (1986)
<https://doi.org/10.1145/16856.16888>
- Stonebraker, Rowe & Hirohama. The implementation of Postgres (1990)
<https://doi.org/10.1109/69.50912>
- Joseph M. Hellerstein. Looking Back at Postgres (2019)
<https://arxiv.org/abs/1901.01973>

THE DESIGN OF POSTGRES

Michael Stonebraker and Lawrence A. Rowe

*Department of Electrical Engineering
and Computer Sciences
University of California
Berkeley, CA 94720*

Abstract

This paper presents the preliminary design of a new database management system, called POSTGRES, that is the successor to the INGRES relational database system. The main design goals of the new system are to:

- 1) provide better support for complex objects,
- 2) provide user extendibility for data types, operators and access methods,
- 3) provide facilities for active databases (i.e., alerters and triggers) and inferencing including forward- and backward-chaining,
- 4) simplify the DBMS code for crash recovery,
- 5) produce a design that can take advantage of optical disks, workstations composed of multiple tightly-coupled processors, and custom designed VLSI chips, and
- 6) make as few changes as possible (preferably none) to the relational model.

Чего хотел Стоунбрейкер

- Сложные объекты (неатомарные типы данных)
- Расширяемость типов данных, методов доступа, операторов.
- Элементы «Активной БД» (триггеры, правила, *alerters*)
- Развитие языка (PostQUEL)
- Рекурсивные запросы
- Компиляция запросов (использование планов)
- Process per user

Чего ещё хотел Стоунбрейкер

- VACUUM – перенос старых данных на оптический диск
- 64-битный transaction id (!)
- Исторические данные (версионность, снапшот)

Tuples will have all non-null fields stored adjacently in a physical record. Moreover, there will be a tuple prefix containing the following extra fields:

IID	: immutable id of this tuple
tmin	: the timestamp at which this tuple becomes valid
BXID	: the transaction identifier that assigned tmin
tmax	: the timestamp at which this tuple ceases to be valid
EXID	: the transaction identifier that assigned tmax
v-IID	: the immutable id of a tuple in this or some other version
descriptor	: descriptor on the front of a tuple

The descriptor contains the offset at which each non-null field starts, and is similar to the data structure attached to System R tuples [ASTR76]. The first transaction identifier and timestamp

И чего ещё хотел Стоунбрейкер

- Восстановление после сбоев дисков
- Лог транзакций с циклическим «ускорителем» в памяти

1. Supporting ADTs in a Database System
 - a. Complex Objects (i.e., nested or non-first-normal form data)*
 - b. User-Defined Abstract Data Types and Functions*
 - c. Extensible Access Methods for New Data Types*
 - d. Optimizer Handling of Queries with Expensive UDFs
2. Active Databases and Rules Systems (Triggers, Alerts)*
 - a. Rules implemented as query rewrites[†]
 - b. Rules implemented as record-level triggers[†]
3. Log-centric Storage and Recovery
 - a. Reduced-complexity recovery code by treating the log as data,* using non-volatile memory for commit status[†]
 - b. No-overwrite storage and time travel queries[†]
4. Support for querying data on new deep storage technologies, notably optical disks*
5. Support for multiprocessors or custom processors*
6. Support for a variety of language models
 - a. Minimal changes to the relational model and support for declarative queries*
 - b. Exposure of "fast path" access to internal APIs, bypassing the query language[†]
 - c. Multi-lingual support[†]

Figure 1: Postgres features first mentioned in the 1986 paper* and the 1991 paper[†].

От Postgres к PostgreSQL

- Postgres пошел в релиз в 1989 г.
- Andrew Yu, Jolly Chen, 1994-95г: Postgres95 (поддержка SQL)
- Тогда же: Postgres стал СПО.
- Вадим Михеев начал работать над MVCC
- 1996: PostgreSQL (6я версия вышла в начале 1997)



Олег выходит на тропу

From megera@sai.msu.su Mon Sep 11 12:06:07 1995
Date: Mon, 11 Sep 1995 12:06:06 +0400 (MSK DST)
From: "O.Bartunov" <megera@sai.msu.su>
X-Sender: megera@ra
To: postgres <postgres95@nobozo.cs.berkeley.edu>
Subject: Q: Friendly interface for Postgres95
Message-ID: <pine.sv4.3.91.950911113736.16173a-100000@ra>
MIME-Version: 1.0
Content-Type: TEXT/PLAIN; charset=US-ASCII
Status: 0
X-Status:

Hi,
now when postgres95 v.1.0 installed and passed all tests
(without regex :-())

NUMBER OF SHARES 1400
CERTIFICATE NUMBER A-10
CLASS A
DATE OF ISSUE SEP 30, 2000

CORPORATION POSTGRESQL, INC
REGISTERED HOLDER OLEG BARTUNOV
TRANSFER (OR ALLOTMENT) FROM Treasury

CERTIFICATE RECEIVED
DATE _____ 20____
Signature _____

<u>A-10</u> Share Certificate Number	<u>A</u> Class of Shares	<u>1400</u> Number of shares
--	-----------------------------	---------------------------------

THIS CERTIFIES THAT OLEG BARTUNOV
is the registered holder of the above described fully paid shares in the capital of

POSTGRESQL, INC.
Incorporated under the "Canada Business Corporations Act"

IN WITNESS WHEREOF the Corporation has caused this
Certificate to be signed by its duly authorized officer(s) this
day of _____, 20____

[Signature]

Внутри Postgres95

```
mege@ra:~/TT_DB$ ls -l
total 36
-rw-r--r-- 1 mege mege 1024 июн 16 1996 access_table.ind
-rw-r--r-- 1 mege mege 3072 июн 16 1996 access_table.rec
-rw-r--r-- 1 mege mege 2048 июн 16 1996 file_object_map.ind
-rw-r--r-- 1 mege mege 2048 июн 16 1996 file_object_map.rec
-rw-r--r-- 1 mege mege 2048 июн 16 1996 file_table.ind
-rw-r--r-- 1 mege mege 3072 июн 16 1996 file_table.rec
-rw-r--r-- 1 mege mege      0 июн 16 1996 file_table.var
-rw-r--r-- 1 mege mege 1024 окт  7 1996 property_table.ind
-rw-r--r-- 1 mege mege 3072 окт  7 1996 property_table.rec
-rw-r--r-- 1 mege mege 1024 окт  7 1996 property_table.var
mege@ra:~/TT_DB$
```

Первый патч Олега

```
commit 5b1311acfbdb6b84dbb84975240914214c51fcb48
```

```
Author: Marc G. Fournier <scrappy@hub.org>
```

```
Date: Wed Apr 2 18:13:47 1997 +0000
```

```
From: Oleg Bartunov <oleg@sai.msu.su>
```

```
Subject: [HACKERS] locale patches !
```

```
Hi there,
```

```
here are little patches to get Postgres 6.1 works with locale stuff.  
This is a patch against 970402.tar.gz, there are no problem to apply them  
by hand to 6.0 release. Collate stuff tested about 1-2 months in real  
working database but I'm sure there must be no problem. US hackers  
could vote against locale implementation ( locale for sure will affect to  
speed of postgres ), so I introduce variable USE_LOCALE which  
controls locale stuff. Non-US users now could use ~* operator  
for searching and <order by> for strings with nation alphabet.  
Please, don't forget, as I did first time, to set environment variable  
LC_CTYPE and LC_COLLATE because backend get locale information from them.  
I start postmaster from a little script, assuming that shell is Bash shell  
it looks like:
```

Рождение PostgreSQL в письмах

```
From: "Vadim B. Mikheev" <vadim@sable.krasnoyarsk.su>  
Date: Wed, 18 Sep 1996 15:04:22 +0800  
Subject: Re: [PG95-DEV] postgres2.0
```

Marc G. Fournier wrote:

[...]

> How about PostgresV6?

And it's good name. But Ingres & Postgres are different in language (not only), and Postgres & Postgres'95 too, so why not to change prefix now again ?

> Please...though...**just not Postgres/SQL**...it just **doesn't slide off the tongue nicely** :(

Vote against too.

Получение имени

commit 9b41da6ce48e3bed6730faa6347a5461175cff83

Author: Bruce Momjian <bruce@momjian.us>

Date: Wed Dec 11 00:28:15 1996 +0000

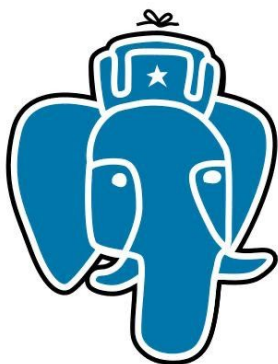
Rename postgres95 to PostgreSQL.
Add comment for SELECT NULL

Русский след...

Slonik.gif,
Даниил Лундин,
Дмитрий Самарсов



<https://obartunov.livejournal.com/186860.html>



WayBackMachine			http://www.ca.postgresql.org/images/	Go
1 captures				
17 Dec 02 - 17 Dec 02				
[IMG]	s3_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s3_on.gif	21-Sep-2001 10:28	1k	
[IMG]	s4_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s4_on.gif	21-Sep-2001 10:28	1k	
[IMG]	s5_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s5_on.gif	21-Sep-2001 10:28	1k	
[IMG]	s6_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s6_on.gif	21-Sep-2001 10:28	1k	
[IMG]	s7_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s7_on.gif	21-Sep-2001 10:28	1k	
[IMG]	s8_off.gif	21-Sep-2001 10:28	1k	
[IMG]	s8_on.gif	21-Sep-2001 10:28	1k	
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[IMG]	slonik.gif	21-Sep-2001 10:28	6k	
[IMG]	so1.gif	21-Sep-2001 10:28	1k	
[IMG]	so2.gif	21-Sep-2001 10:28	1k	
[IMG]	so3.gif	21-Sep-2001 10:28	1k	
[IMG]	so4.gif	21-Sep-2001 10:28	1k	
[IMG]	so5.gif	21-Sep-2001 10:28	1k	
[IMG]	spacer.gif	21-Sep-2001 10:28	1k	
[IMG]	sqlephant.gif	21-Sep-2001 10:28	16k	
[IMG]	ss.gif	21-Sep-2001 10:28	1k	
[IMG]	tbkg.gif	21-Sep-2001 10:28	1k	
[IMG]	title.gif	21-Sep-2001 10:28	2k	
[IMG]	top.gif	21-Sep-2001 10:28	10k	
[IMG]	verh1.gif	21-Sep-2001 10:28	1k	
[IMG]	verh2.gif	21-Sep-2001 10:28	5k	
[IMG]	verh3.gif	21-Sep-2001 10:28	1k	
[IMG]	zaglushka.gif	21-Sep-2001 10:28	1k	

Этапы развития

- 1996 — Стабилизация работы
- 1997 (6.1) — Интернационализация +World
- 2005 (8)
- 2010 (9) — Встроенная репликация +Enterprise users
- 2014 (9.4) — jsonb +NoSQL users
- 2016 (9.6) — Parallel Query +OLAP users
- 2017 (10) — Logical Replication, Declarative Partitioning
- 2018 (11) — JIT
- 2019 (12) — Pluggable storage, Jsonpath (SQL/JSON)
- 2022 (15) — JSON_TABLE,...(SQL/JSON), MERGE
- 2023 (16) — Pluggable TOAST, Fast Jsonb

MVCC

- Выпилить историческое хранилище Стоунбрейкера
- Уровни изоляции транзакций
- Сделал Вадим Михеев. (См).

Date: Mon, 09 Aug 1999 16:08:19 +0800
From: Vadim Mikheev
To: Oleg Bartunov
Subject: Re: indices grow !

-----skipped-----

А что его понимать-то! -:)

Основное что надо помнить:

Запрос (те Query - то что читает записи из базы используя Seq/Index scans и отбирает их в соответствии с условиями в WHERE используя joins, subselects etc) видит (те возвращает) только те записи, который были живы в момент старта запроса(READ COMMITTED)/транзакции(SERIALIZABLE).

Всё остальное лишь производное -:)

WAL

- Сделал Вадим Михеев.
- Основывается на **ARIES** (1992), IBM (С.Mohan).
 - Запись вначале в WAL, на надёжный сторадж
 - Проигрывание истории после крэша
 - Rollback тоже, конечно же, пишется в WAL
 - Позже это пригодилось для репликации.

```
commit 47937403676d913c0e740eec6b85113865c6c8ab
Author: Vadim B. Mikheev <vadim4o@yahoo.com>
Date:   Wed Oct 6 21:58:18 1999 +0000
```

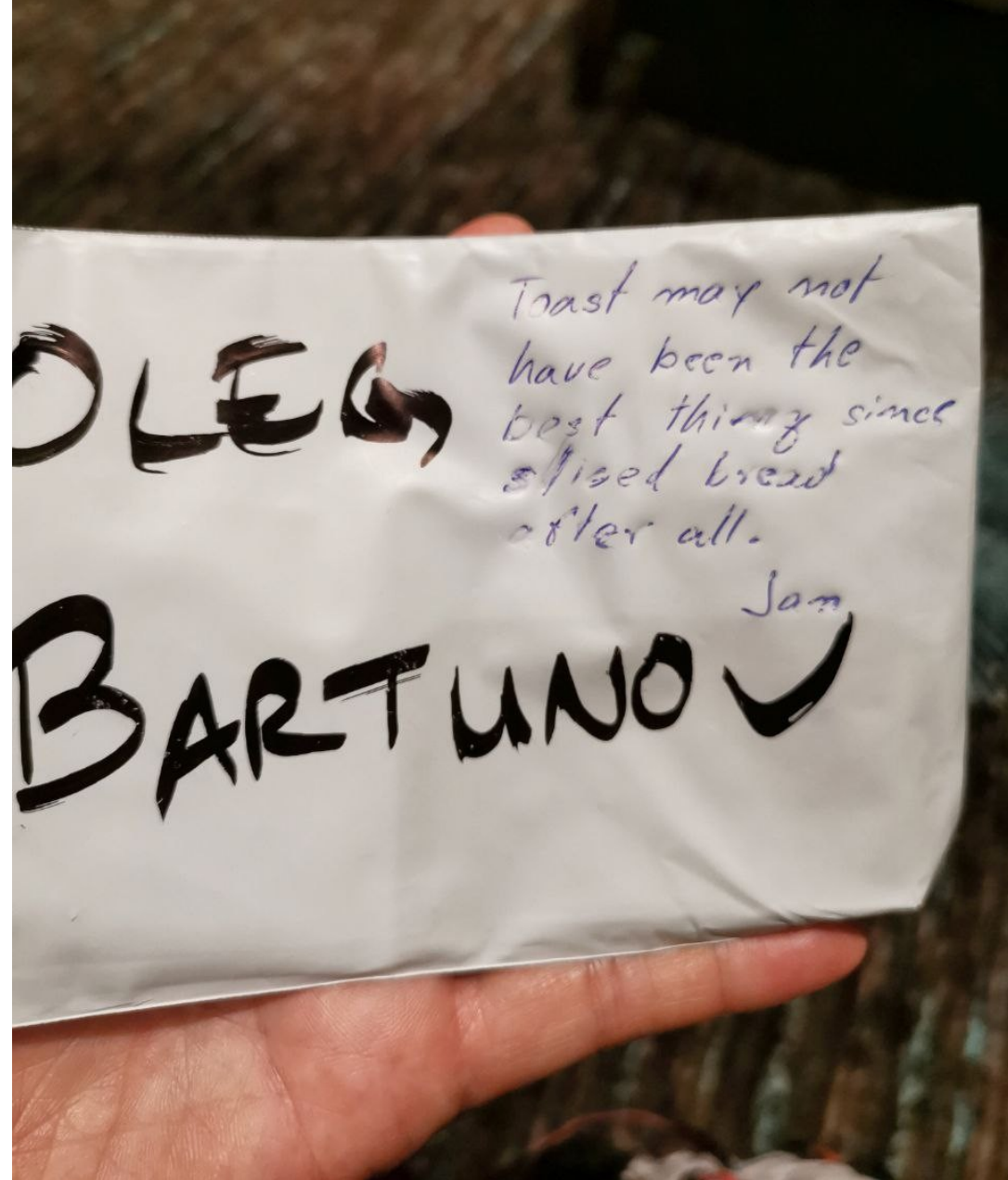
XLOG (also known as WAL -:)) Bootstrap/Startup/Shutdown.
First step in cleaning up backend initialization code.
Fix for FATAL: now FATAL is ERROR + exit.

Дальнейшее развитие MVCC

- Проблема Write Amplification
 - HOT
- Проблема определения видимости версий
 - hint bits
 - ProcArray vs CSN

История TOAST

- Был написан Jan Wieck, Dec 1999
- С тех пор не менялся, разве что:
 - Частичное чтение (Partial TOAST decompression)
 - Идёт обсуждение Pluggable TOASTers.



История GiST

- Разработан в Berkeley для PostgreSQL командой проф Joe Hellerstein
<http://gist.cs.berkeley.edu/>
- Затем полностью переписан О.Бартуновым и Ф.Сигаевым в начале XXI в.
- GiST — обобщенная структура, поверх реализуются разные типы деревьев. RD-дерево для поиска в множествах. ←

Author: This extraction from an e-mail sent by **Eugene Selkov Jr.** contains good information on GiST. Hopefully we will learn more in the future and update this information. - thomas 1998-03-01

Well, I can't say I quite understand what's going on, but at least I (almost) succeeded in porting GiST examples to linux. The GiST access method is already in the postgres tree (src/backend/access/gist).

Examples at Berkeley come with an overview of the methods and demonstrate spatial index mechanisms for 2D boxes, polygons, integer intervals and text (see also **GiST at Berkeley**). In the box example, we are supposed to see a performance gain when using the GiST index; it did work for me but I do not have a reasonably large collection of boxes to check that. Other examples also worked, except polygons: I got an error doing

```
test=> create index pix on polytmp
test-> using gist (p:box gist_poly_ops) with (islossy);
ERROR:  cannot open pix
```

Sven Helmer. Index Structures for Databases Containing Data Items with Set-valued Attributes (1997)

Hellerstein о том, как Олег и Федор переписали GiST

nity made significant modifications and improvements to the core of the system as well, from the optimizer to the access methods and the core transaction and storage system. Since the mid-1990s, very few of the PostgreSQL internals came out of the academic group at Berkeley—the last contribution may have been my GiST implementation in the latter half of the 1990s—but even that was rewritten and cleaned up substantially by open-source volunteers (from Russia, in that case). The open source community around PostgreSQL deserves enormous credit for running a disciplined process that has soldiered on over decades to produce a remarkably high-impact and long-running project.

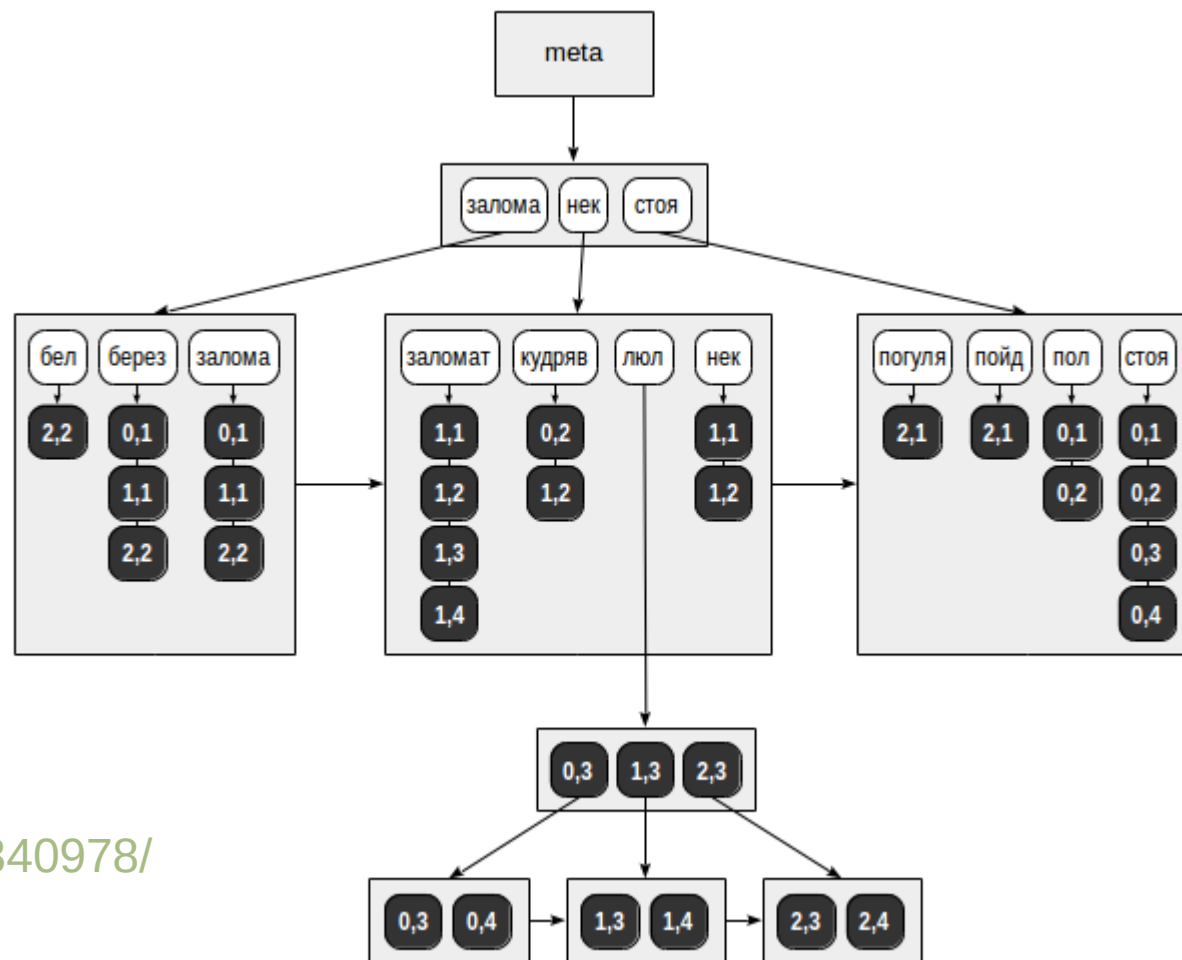
Довольные авторы GiST



- Проблема GiST (точнее, проблема superimposed сигнатур) — переполнение этих сигнатур.

GIN-индексы

- Обобщенный обратный индекс. Сигаев, Бартунов (8.4, 2008)
- Ключ => B-дерево из xid



<https://habr.com/ru/company/postgrespro/blog/340978/>

История репликации

- Transaction log shipping (Mammoth Replicator, Slony)
- Proxy-based replication (PgPool)

Replication is not a single solution for a single problem; it is several solutions for a wide array of different problems. No one replication tool will ever be the "default" replication for PostgreSQL

Josh Berkus, 2004

- WAL-based replication: Hans-Jürgen Schönig (idea, 2005).
Релиз 9.0 (2010)
- Logical replication (pglogical, 9.4, 2015)

История репликации (2)

- Postgres XC
- Postgres XL
- BDR
- Postgres Pro Multimaster
-

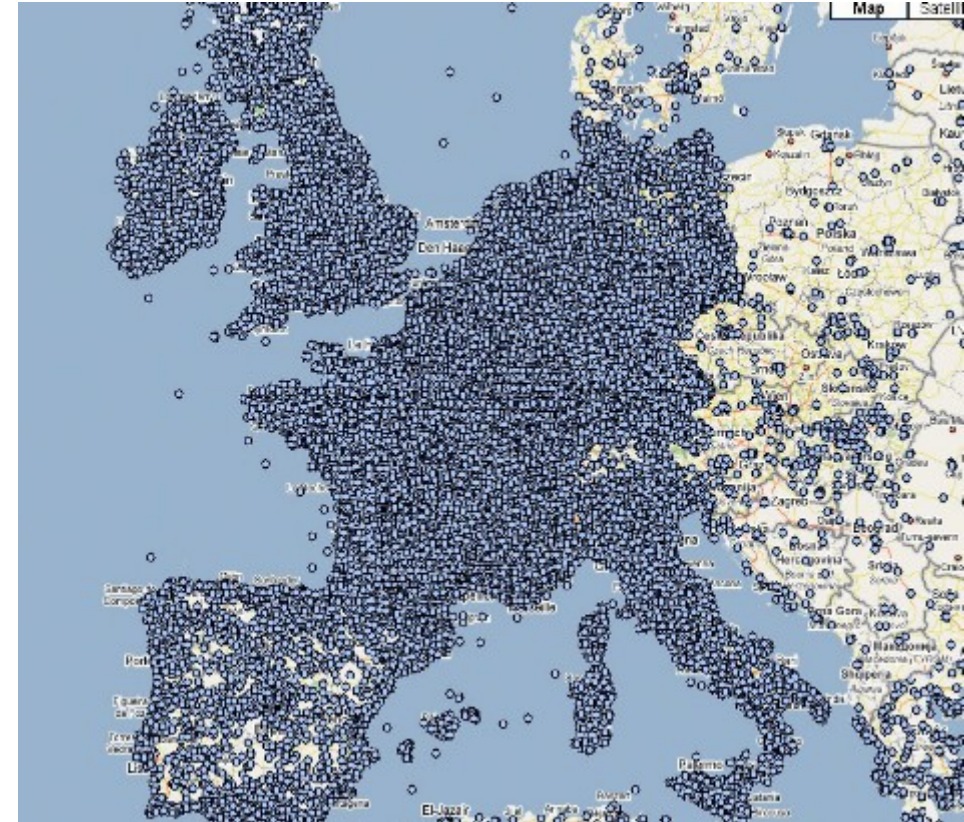
История JSON(,B!)

- Intarray, GiST (2000, Бартунов, Сигаев, Rambler)
- HStore с 2003 (с 2006 в PostgreSQL, Бартунов, Сигаев)
- JSON как текст 9.2
- JSONB (9.4) и GIN-индексы (Бартунов, Коротков, Сигаев)
- SQL/JSON (12... ? Бартунов, Коротков, Глухов)
- GSON ?

KNN

- 2005–2010. GiST, PostGIS
- 2020. SP-GiST (А. Лебедев)
- B-Tree

Бартунов, Сигаев, Глухов



История JSON(,B!)

- Intarray, GiST (2000, Бартунов, Сигаев, Rambler)
- HStore с 2003 (с 2006 в PostgreSQL, Бартунов, Сигаев)
- JSON как текст 9.2
- JSONB (9.4) и GIN-индексы (Бартунов, Коротков, Сигаев)
- SQL/JSON (12... ? Бартунов, Коротков, Глухов)
- GSON ?

История полнотекстового поиска

- SQL-based
- OpenFTS (2000..2009, 7.4) - поиск на базе GiST (Сигаев, Бартунов)
- Tsearch
- Tsearch2
- Встроенный поиск
- GIN-индекс

История резервного копирования

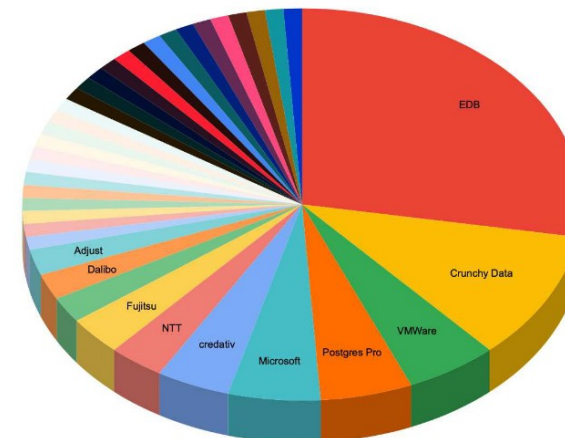
- pg_dump
- pg_basebackup + WAL archiving
- PITR
- pg_probackup + PTrack : инкрементальный Backup

Расширяемость

- Типы данных (tsvector, json, xml, pgsphere, PostGIS...)
- Индексы (GiST, SP-GiST, GIN, BRIN)
- Языки программирования (PL/Perl, Python, V8, Java).
- Табличные методы доступа (Heap...)
- TOAST (в процессе...)
-

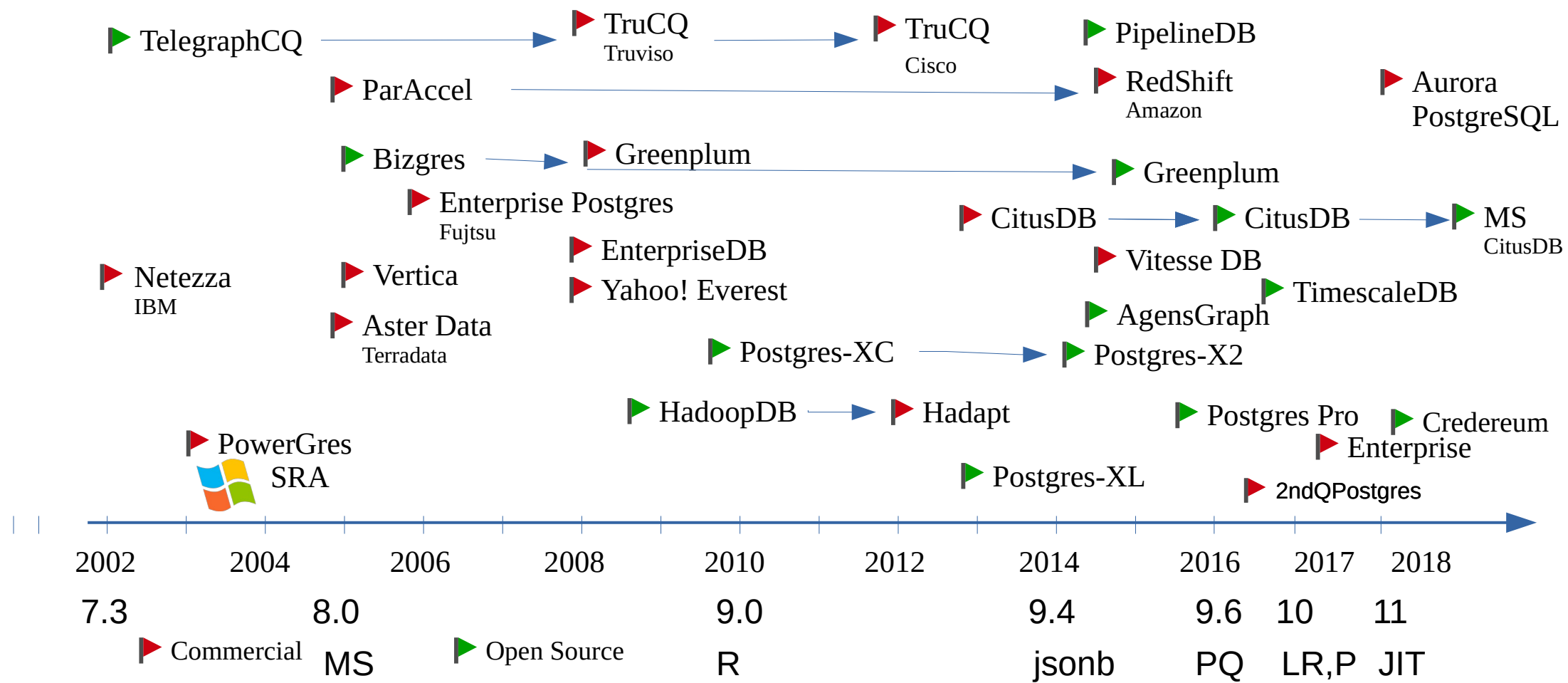
Сообщество

- Пермиссивная лицензия
 - Форки ←
 - Бэкпорт →
- Эволюция
 - от гиков до корпораций
- Баланс между developer-driven и customer-driven



EDB Crunchy Data VMware Postgres Pro Microsoft credativ NTT Fujitsu
PeopleDoc Dalibo Adjust Yandex University of Cambridge True Software Swarm64
SRA OSS Rakuten Postgres Experts Playtech Paragon
Open Standards Promotion Center, Japan Ongres MongoDB Minsait iovation ILande
Huawei HeteroDB Google Facebook Deriv.com Cybertec
conova communications GmbH Brandwatch Basefarm AS Appen Amazon Afiliias

Форки





Спрашивайте Postgres Pro и приходите на PgConf.RU!



Оценить доклад



Saint HighLoad⁺⁺

НТИКО